

EXECUTIVE SUMMARY

BRIEF INTRODUCTION ON THE DATASET

The dataset reveals 1200 rows and 14 columns of sales data which include Order-ID, Date, Customer-ID, Product, Quantity, Unit Price, Shipping Address, Payment Method, Order Status, Tracking Number, Items in cart, Coupon code, Referral source and Total Price.

The dataset includes 6 text datatype, 3 alphanumeric datatype, 1 date, and 4 Numeric datatypes.

The datatype is about a retail company's sales data, that focuses on sales of 6 products, which use online platforms for its sales channel. The retail company give room for discount with the use of coupon code for some of its customers. The company also give room for various delivery channels.

DATA CLEANING

During the cleaning of this dataset, I checked the number of rows and columns, I looked to understand what the dataset is all about. Then I proceed to check for duplicates, spelling errors, null values and incorrect datatype. I trimmed some columns should in case there are white spaces. I discovered null values in the Coupon code column. I then replace the null values with "No Coupon" reason been, my observation of real-world data shows that some customers do not have access to promo or discount, due to time constraints of some promo or discount, the conditions attached to the promo or discount or because of no access to promo information.

SUMMARY STATISTICS.

I did some summary statistics on my dataset (particularly on the Total Price Column.

I checked the Minimum value, Maximum value, Average of Total Price, the count of Customer ID, Median, 25th and 75th quartile, to determine Inter Quartile Range (IQR), then I check the lower and upper bound to determine if there is an outlier in our Total Price column. I use Conditional statement to test for the presence of outliers.

PIVOT TABLES

I created pivot tables to summaries my data to further explore the data for analysis.

The Key performance indicators (KPI's) include:

- Total Count of Order = 1200
- Total Count of Customer = 1200
- Total Item in cart = 6582

- Total Quantity sold = 3535
- Total Revenue = \$1.3M
- AOV = \$1054

Monthly sales trend – Quantity/ Revenue, with June as the month with the highest sale.

Product by sales Quantity, Chair is the most sold product.

Referral Source by revenue- Instagram was the most used referral source.

Product by sales quantity and revenue – Chair is the most sold product and generated the highest revenue.

Which payment Method is the most used - Cash been the most used payment method.

Coupon code by Revenue and Quantity – Free ship is the highest used coupon code, generating about \$335, 037, with quantity of about 914, followed by No coupon with \$322,401 and 941 quantities sold, “No Coupon” get more quantity sold than but got less revenue than Free ship.

I check the yearly sales trend and was bewildered because, the quantity sold and revenue keep decreasing every year for the year.

RECOMMENDATION

I recommend that the company track the reasons behind cancelled goods, returned goods and work on ways to reduce such occurrence either by reducing the return timeline (maybe 7days is given before, it should be reduced to 3days) for items to be returned and failure to do so means you cannot return the product.

We know that cancelling of order on the cart is inevitable but it can be reduced by giving customers assurance of quality for product bought.

I also recommend that the company should focus more on gift items as promo rather than only discount.

The company should prioritize quality products goods over quantity, timely delivery and normalize consistent value of product.

The company to check the reason for yearly reduction in revenue, factor that affect online sales and put in place measures to tackle the issue.